

JUAN CARLOS ZUÑIGA ALFARO

Web Development Intern

Guanajuato, Mexico | +52 464 206 6001 | juanxx514@gmail.com | [LinkedIn](#) | GitHub: github.com/juan7679 | Portfolio: jczuniga-dev.netlify.app

PROFESSIONAL SUMMARY

Computer Systems Engineering student at Universidad de Guanajuato with hands-on experience building web applications with React, Vue.js, JavaScript, HTML, CSS, and REST APIs. I have worked on collaborative full-stack projects where I developed frontend modules, integrated APIs, implemented CRUD functionality, and worked with Git/GitHub and Docker. I also have experience with Python and computer vision through academic projects using OpenCV and MediaPipe. I have deployed and configured full-stack applications end to end — hosting frontends on Netlify, backends on Render, and managing Firebase Admin SDK credentials and environment configuration for live, publicly accessible demos. Currently looking for an internship where I can continue developing my web development skills while contributing to real-world projects.

TECHNICAL SKILLS

- **Web Development:** HTML, CSS, JavaScript, React, Vue.js, Vite, Responsive Web Design, REST APIs, Axios
- **Tools & Development:** Git, GitHub, Docker, Docker Compose, VS Code
- **Backend & Databases:** Node.js, Express, Firebase Firestore, JWT, Python
- **Deployment & Hosting:** Netlify, Render, Firebase Admin SDK, environment variable management, Git branching workflow (feature/develop/main)
- **Engineering Practices:** Debugging, Troubleshooting, Quality Assurance, Documentation, Software Modularization, API Integration
- **Computer Vision:** OpenCV, MediaPipe, DeepFace, FaceNet

SELECTED PROJECTS

EduControl — Academic Management System | *React, Node.js, Express, Firebase, Docker*

- Developed responsive React interfaces for academic management modules including Teachers, Subjects, and Groups as part of a collaborative full-stack application.
- Implemented CRUD workflows and integrated front-end components with REST APIs using Axios.
- Worked with routing, reusable UI components, authentication flows, and role-based application behavior.
- Tested and debugged front-end functionality and API integration to identify and resolve issues during development.
- Collaborated using Git/GitHub and contributed to project documentation covering architecture, APIs, setup, and deployment.
- Worked with a Dockerized development environment using Docker Compose.

DEMEN ERP — Full-Stack ERP System | *Vue 3, JavaScript, Vite, Node.js, Express, Firebase*

- Deployed a live public demo (frontend on Netlify, backend on Render) with a seeded fictitious dataset and a role-restricted demo account. *Live demo available on request.*
- Developed responsive interfaces and reusable components for an enterprise-style ERP application.
- Implemented front-end modules for clients, suppliers, products, inventory, and product receptions.
- Integrated REST APIs using Axios and implemented client-side routing and state management.
- Built responsive layouts with light/dark theme support and reusable UI components.
- Documented application architecture, API endpoints, setup instructions, security practices, and development workflow.

Hand Gesture Drawing — Real-Time Computer Vision App | *Python, OpenCV, MediaPipe*

- Built a real-time webcam application that translates hand gestures into drawing, pause, erase, and color-selection actions.
- Developed gesture recognition and color-selection modules using MediaPipe hand landmarks.
- Improved pinch detection by replacing a fixed pixel threshold with a hand-size-relative threshold for more consistent interaction across camera distances.

EDUCATION

Universidad de Guanajuato — B.S. in Computer Systems Engineering Expected Graduation: August 2027

Relevant Coursework: Web Development, Software Engineering, Databases, Cloud Computing, Artificial Intelligence, Computer Vision

LANGUAGES

- Spanish — Native
- English — Strong reading and writing proficiency; intermediate speaking and listening